Intermediate Javascript

Document Object Model

Dominating the DOM to add functionality to HTML elements

```
<html>
  <head>
    <title>My Website</title>
  </head>
  <body>
    <button>Alert</button>
  </body>
</html>
```



```html
<html>
  <head>
    <title>My Website</title>
  </head>
  <body>
    <button>Alert</button>
  </body>
</html>
```



index.html index.js styles.css

```
<head>
  <meta charset="utf-8">
  <title>My Website</title>
  <link rel="stylesheet" href="styles.css">

</head>
<body>

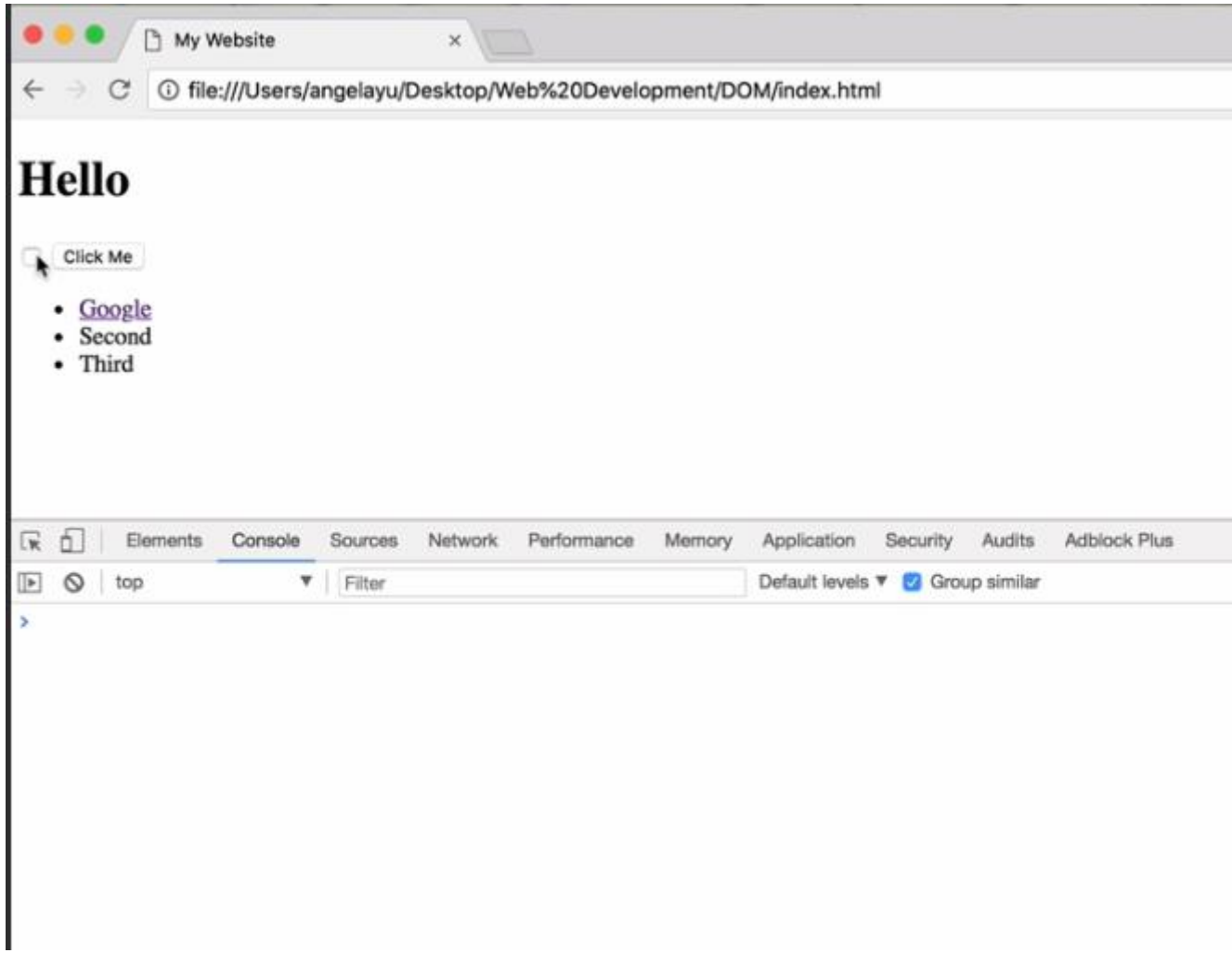
  <h1>Hello</h1>

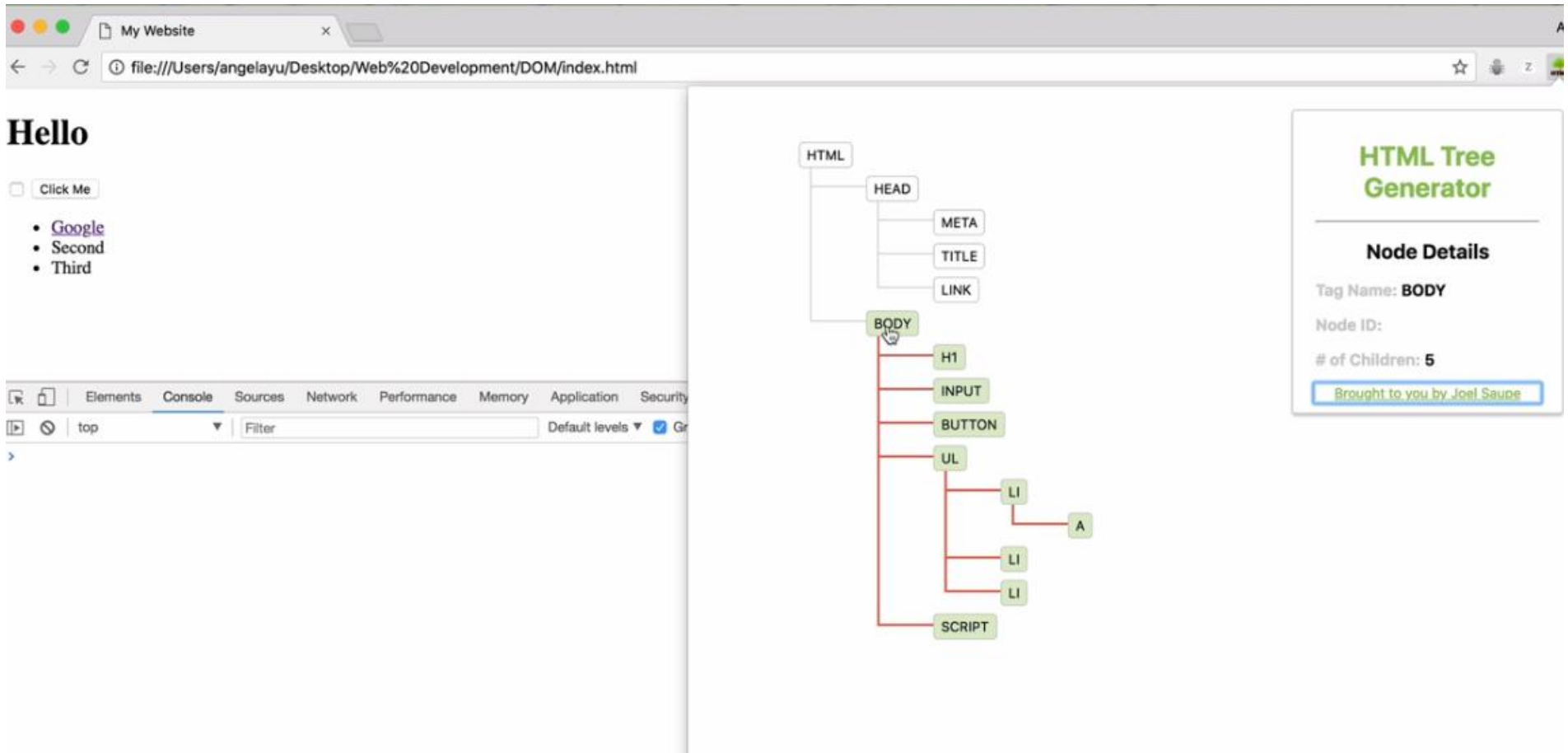
  <input type="checkbox">

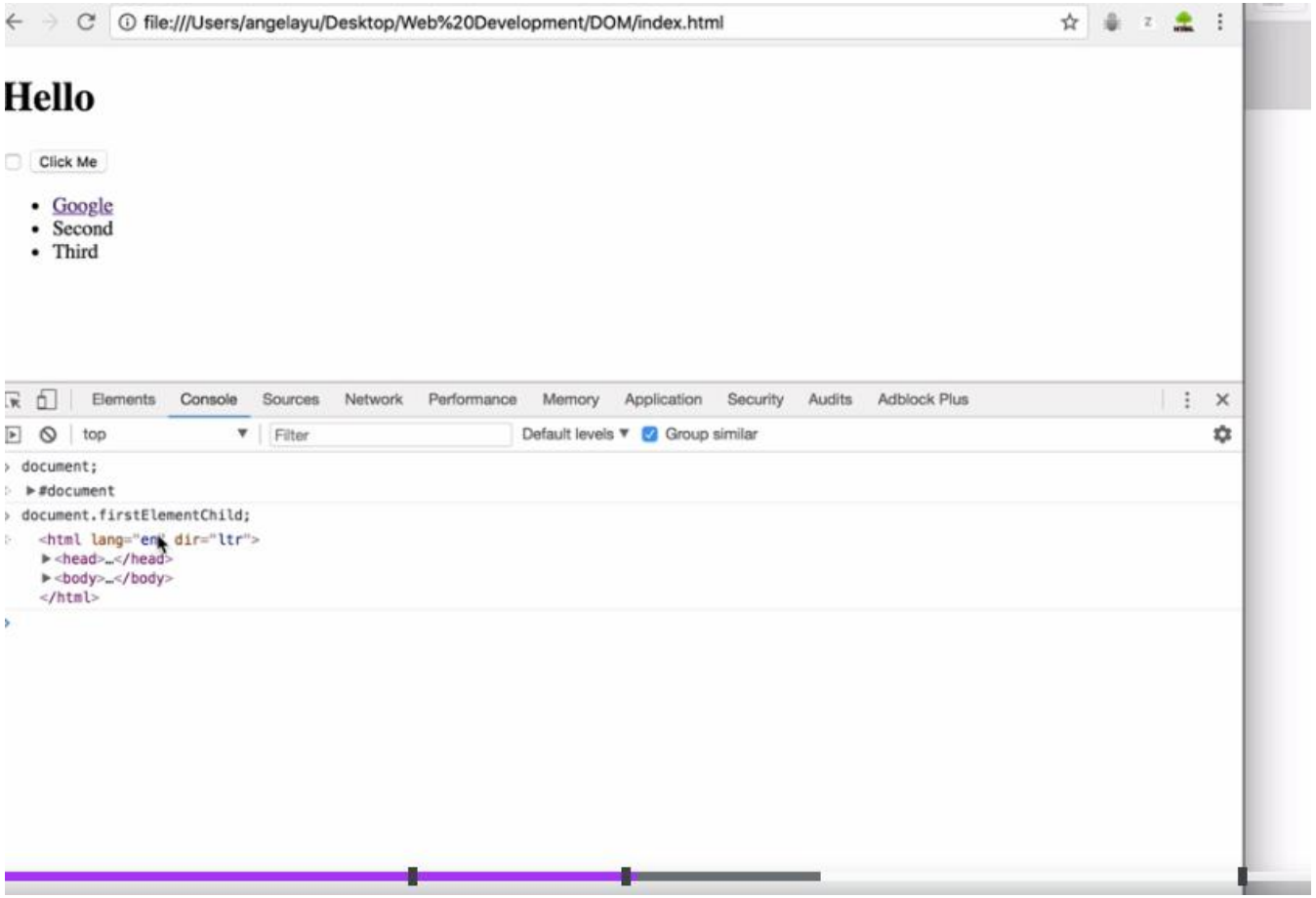
  <button style=":active: color:red;">Click Me</button>

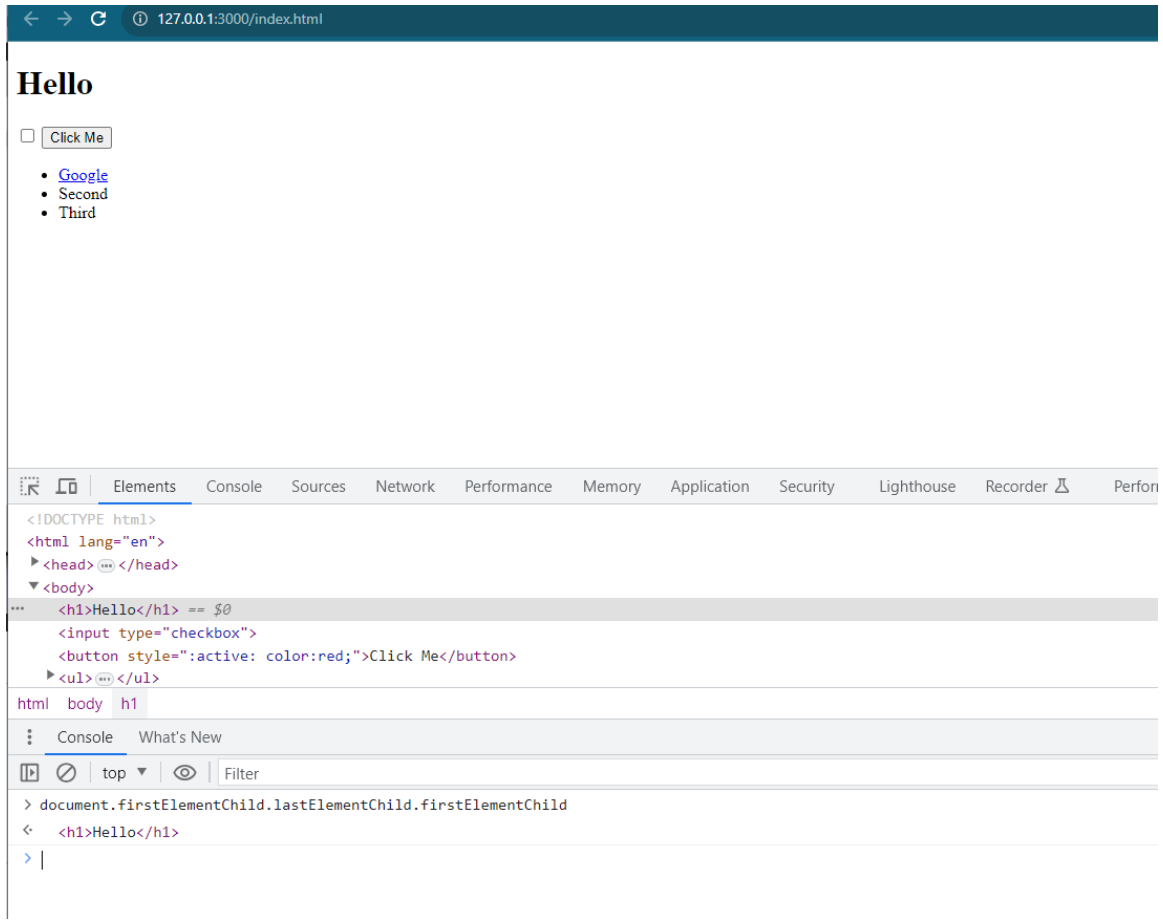
  <ul>
    <li class="list">
      <a href="https://www.google.com">Google</a>
    </li>
    <li class="list">Second</li>
    <li class="list">Third</li>
  </ul>

<script src="index.js" charset="utf-8"></script>
</body>
</html>
```




My Website
file:///Users/angelayu/Desktop/Web%20Development/DOM/index.html
Hello
Click Me
Google
Second
Third
Elements
Console
Sources
Network
Performance
Memory
Application
Security
top
Filter
Default levels
HTML
HEAD
META
TITLE
LINK
BODY
H1
INPUT
BUTTON
UL
LI
A
LI
LI
SCRIPT
HTML Tree Generator
Node Details
Tag Name: BODY
Node ID:
of Children: 5
Brought to you by Joel Saupe


file:///Users/angelayu/Desktop/Web%20Development/DOM/index.html
Hello
Click Me
Google
Second
Third
Elements
Console
Sources
Network
Performance
Memory
Application
Security
Audits
Adblock Plus
top
Filter
Default levels
Group similar
document;
#document
document.firstElementChild;
<html lang="en" dir="ltr">
<head>…</head>
<body>…</body>
</html>


127.0.0.1:3000/index.html
Hello
Click Me
Google
Second
Third
Elements
Console
Sources
Network
Performance
Memory
Application
Security
Lighthouse
Recorder
<!DOCTYPE html>
<html lang="en">
<head> … </head>
<body>
<h1>Hello</h1> == $0
<input type="checkbox">
<button style=":active: color:red;">Click Me</button>
 …
html body h1
Console
What's New
top
Filter
> document.firstElementChild.lastElementChild.firstElementChild
<h1>Hello</h1>

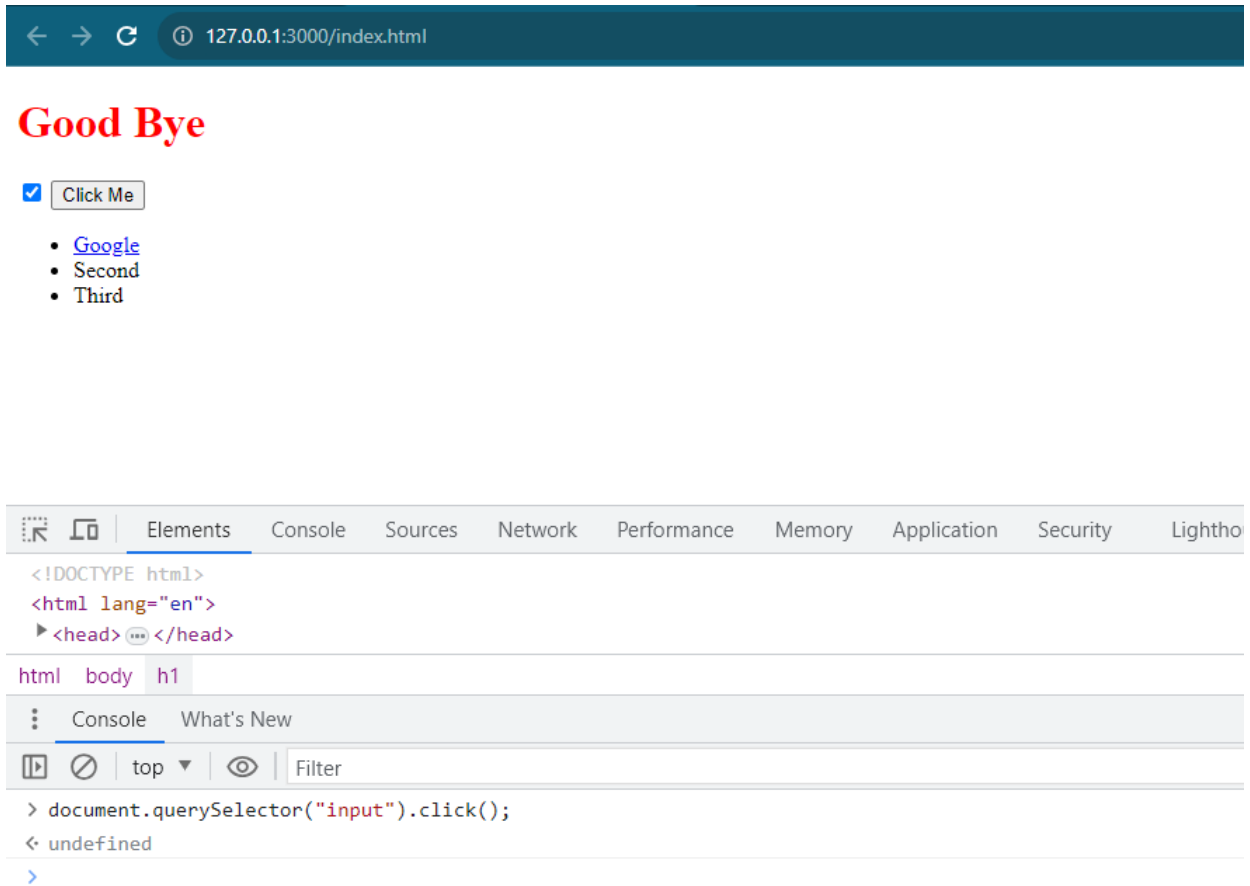


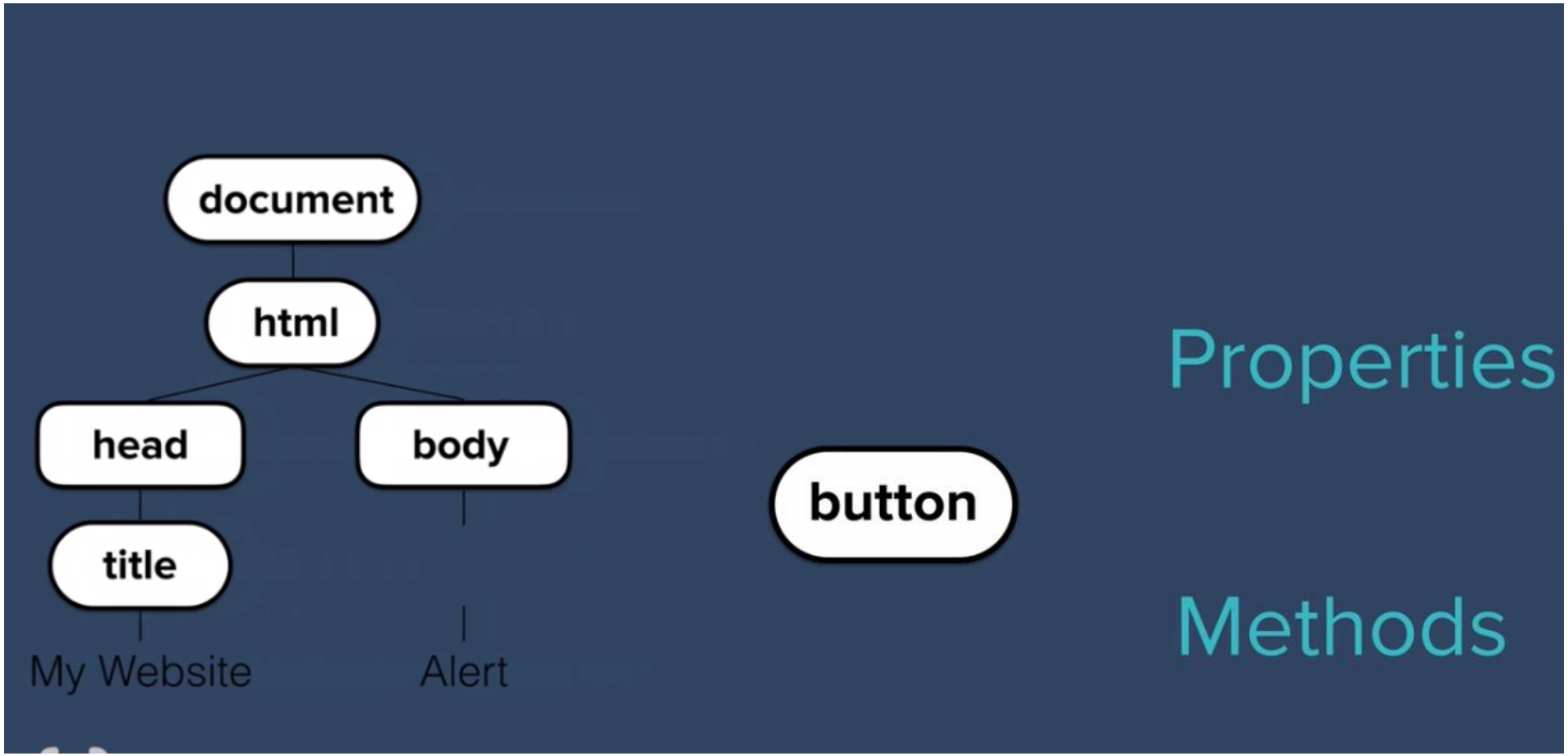





```
var heading = document.firstElementChild.lastElementChild.firstElementChild;

heading.style.color = "red";
```


127.0.0.1:3000/index.html
Good Bye
Click Me
Google
Second
Third
Elements
Console
Sources
Network
Performance
Memory
Application
Security
Lightho
<!DOCTYPE html>
<html lang="en">
<head> </head>
html
body
h1
Console
What's New
top
Filter
> document.querySelector("input").click();
undefined
>


document
html
head
body
title
My Website
Alert
button
Properties
Methods


Properties
• Colour
• Number of seats
• Number of Doors
Methods
• Brake()
• Drive()
• Park()

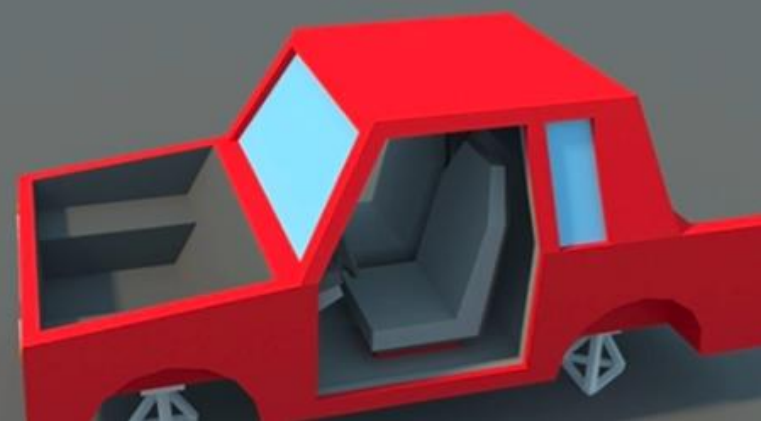
Get Property

```
car.colour; //red
```



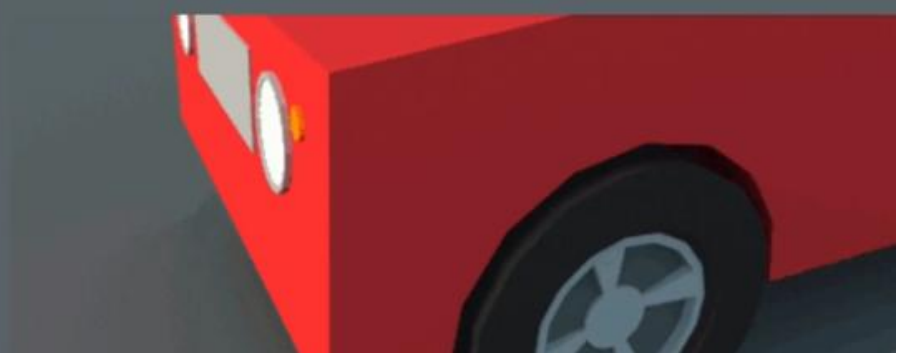
Set Property

```
car.numberOfDoors = 0;
```



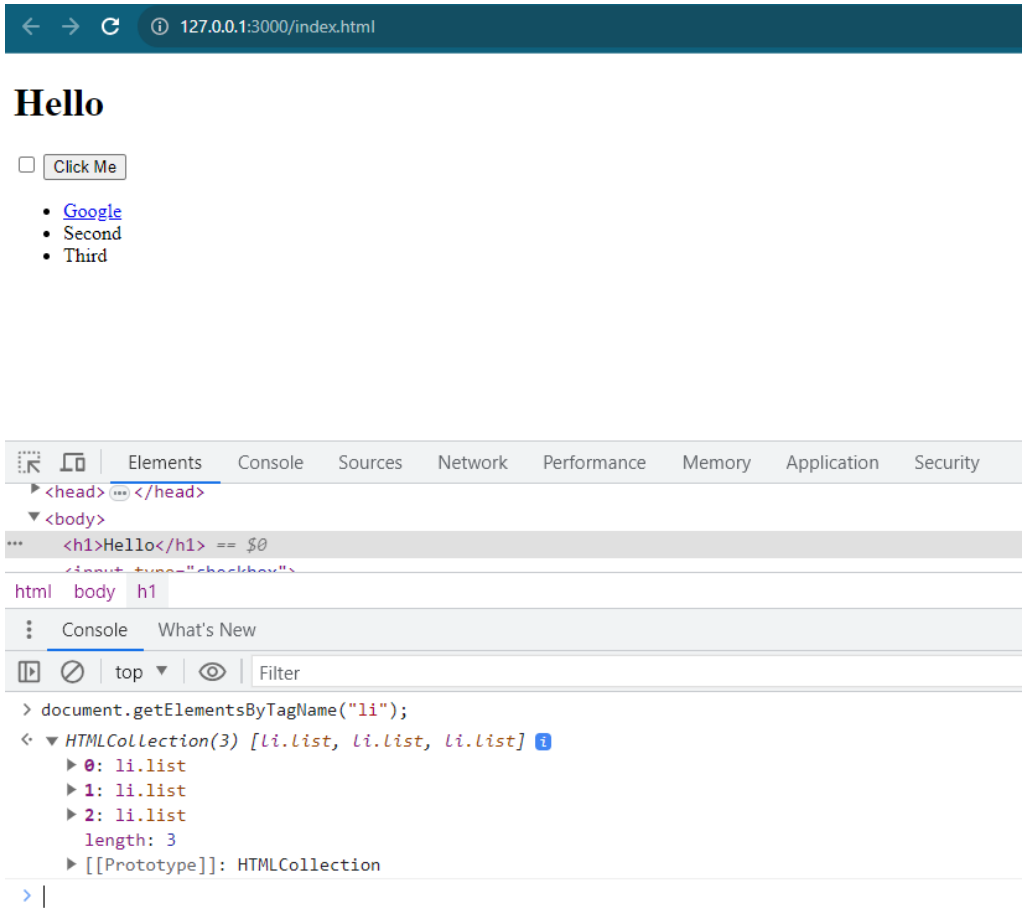
Call Method

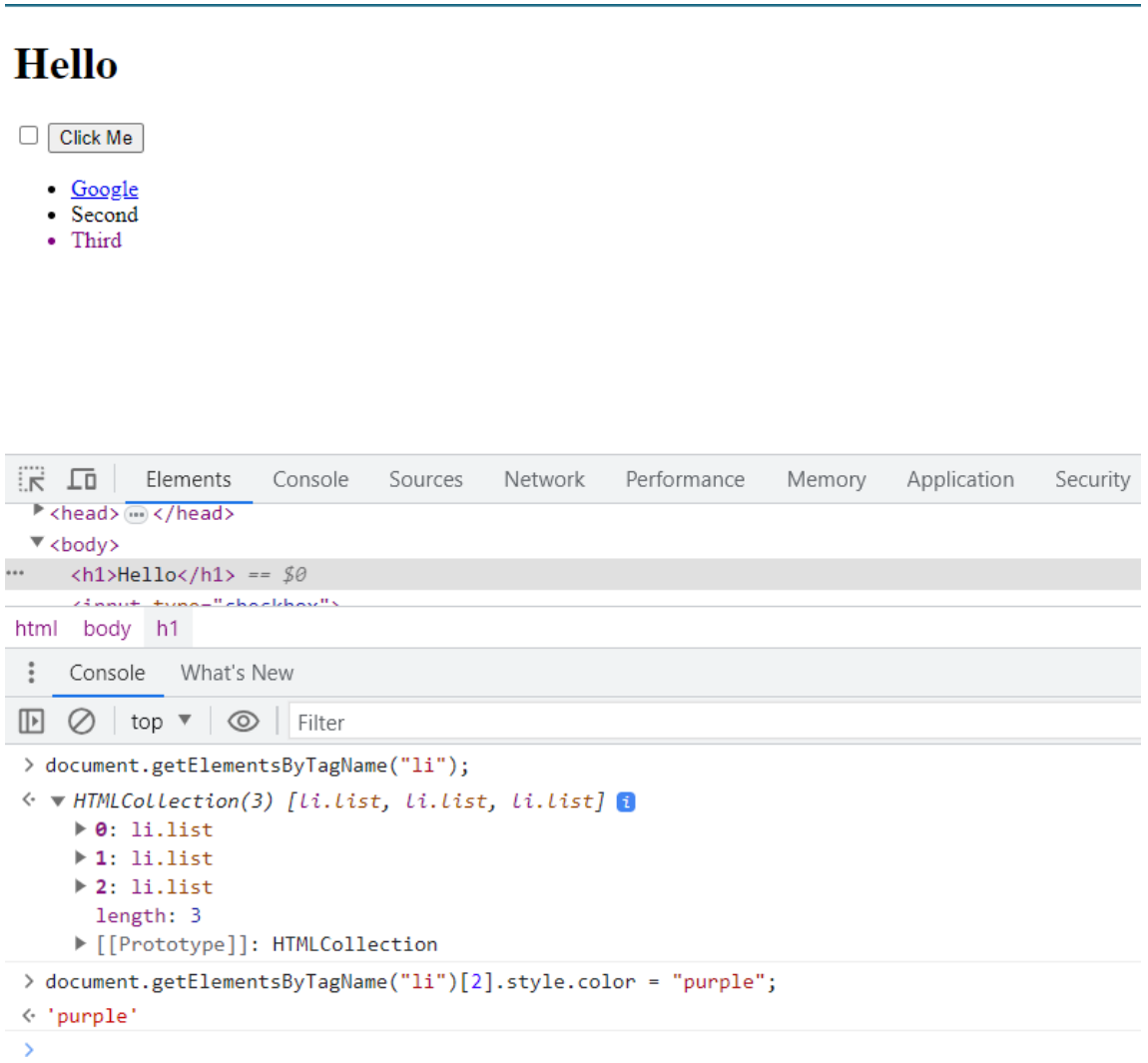
```
car.drive();
```



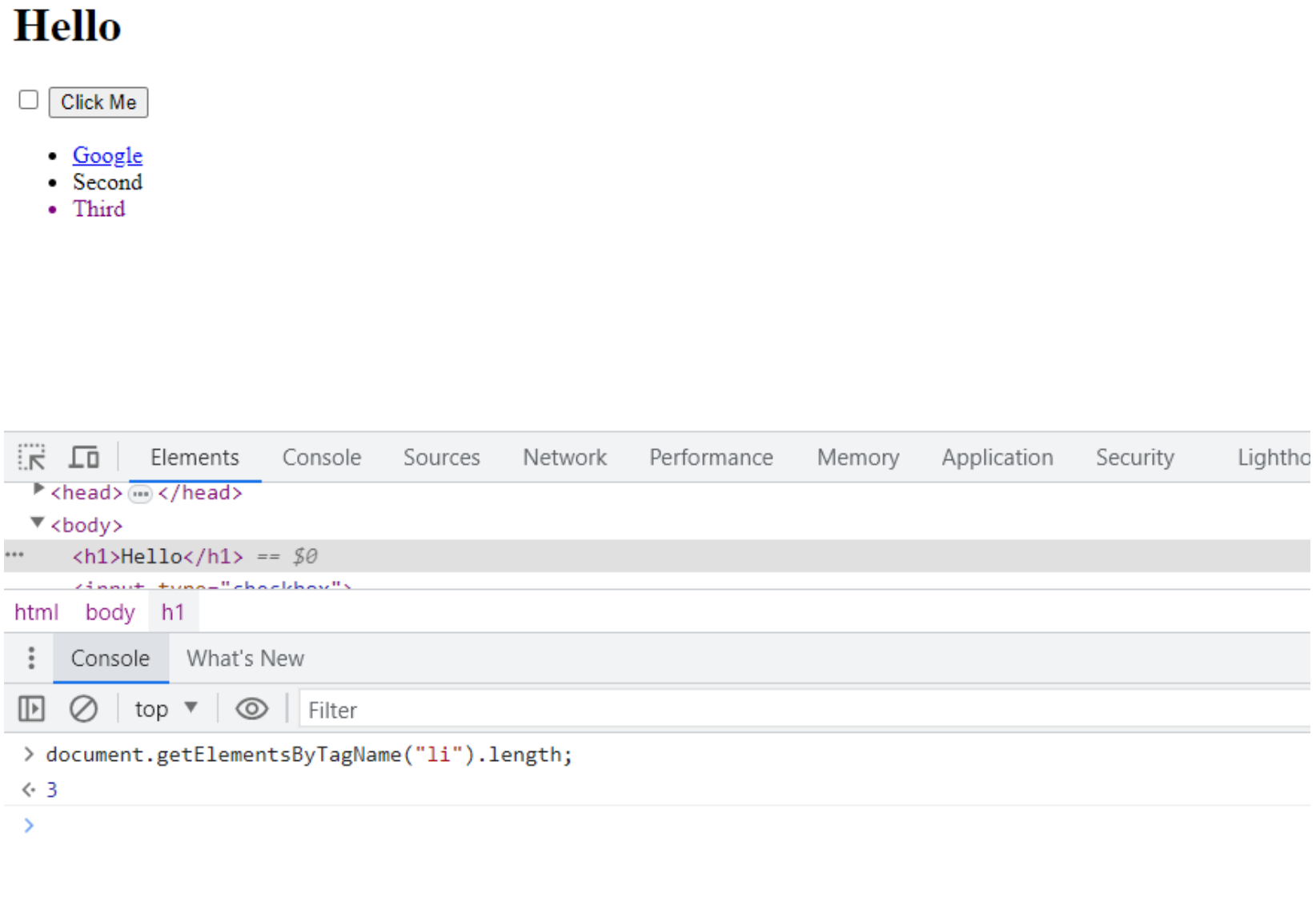

button
Properties
• innerHTML
• style
• firstChild
Methods
• click()
• appendChild()
• setAttribute()

Selecting HTML Elements with Javascript

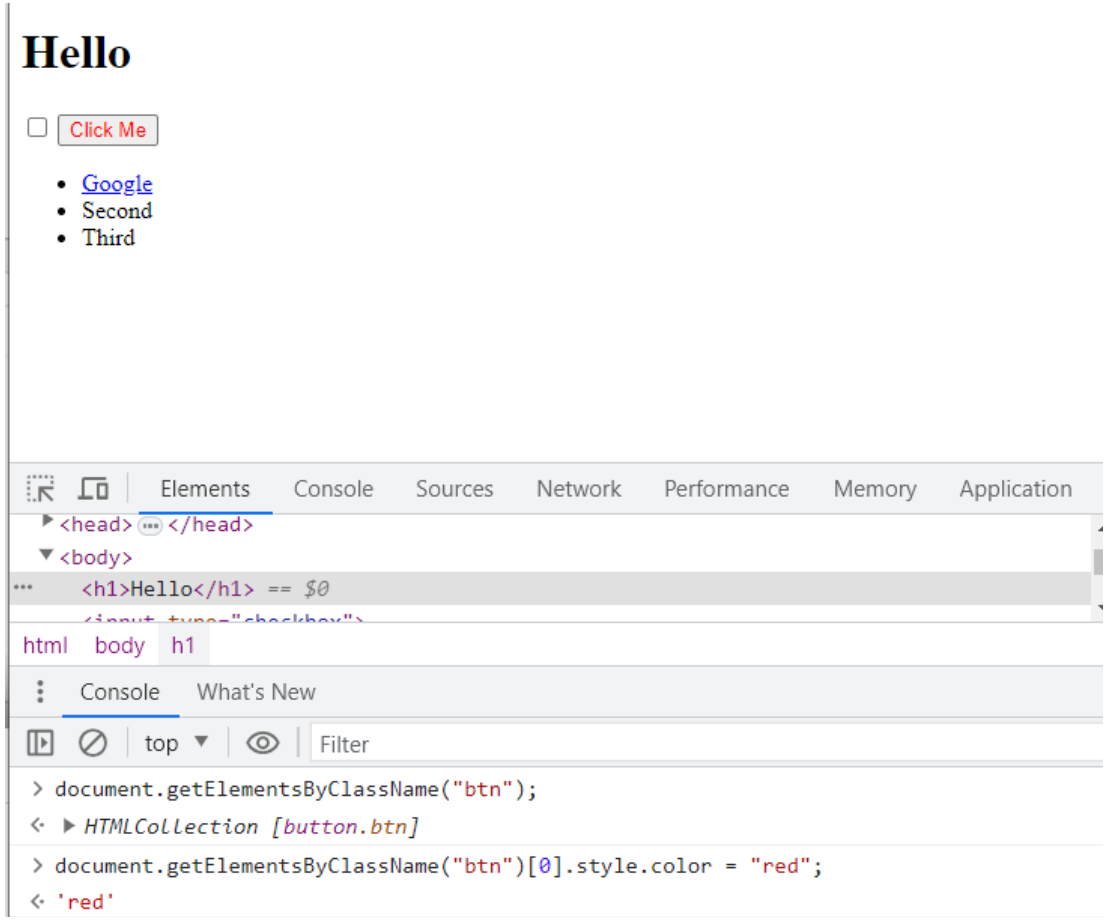




length


Hello
Click Me
Google
Second
Third
Elements
Console
Sources
Network
Performance
Memory
Application
Security
<head> </head>
<body>
<h1>Hello</h1> == $0
html
body
h1
Console
What's New
top
Filter
> document.getElementsByTagName("li").length;
< 3
>

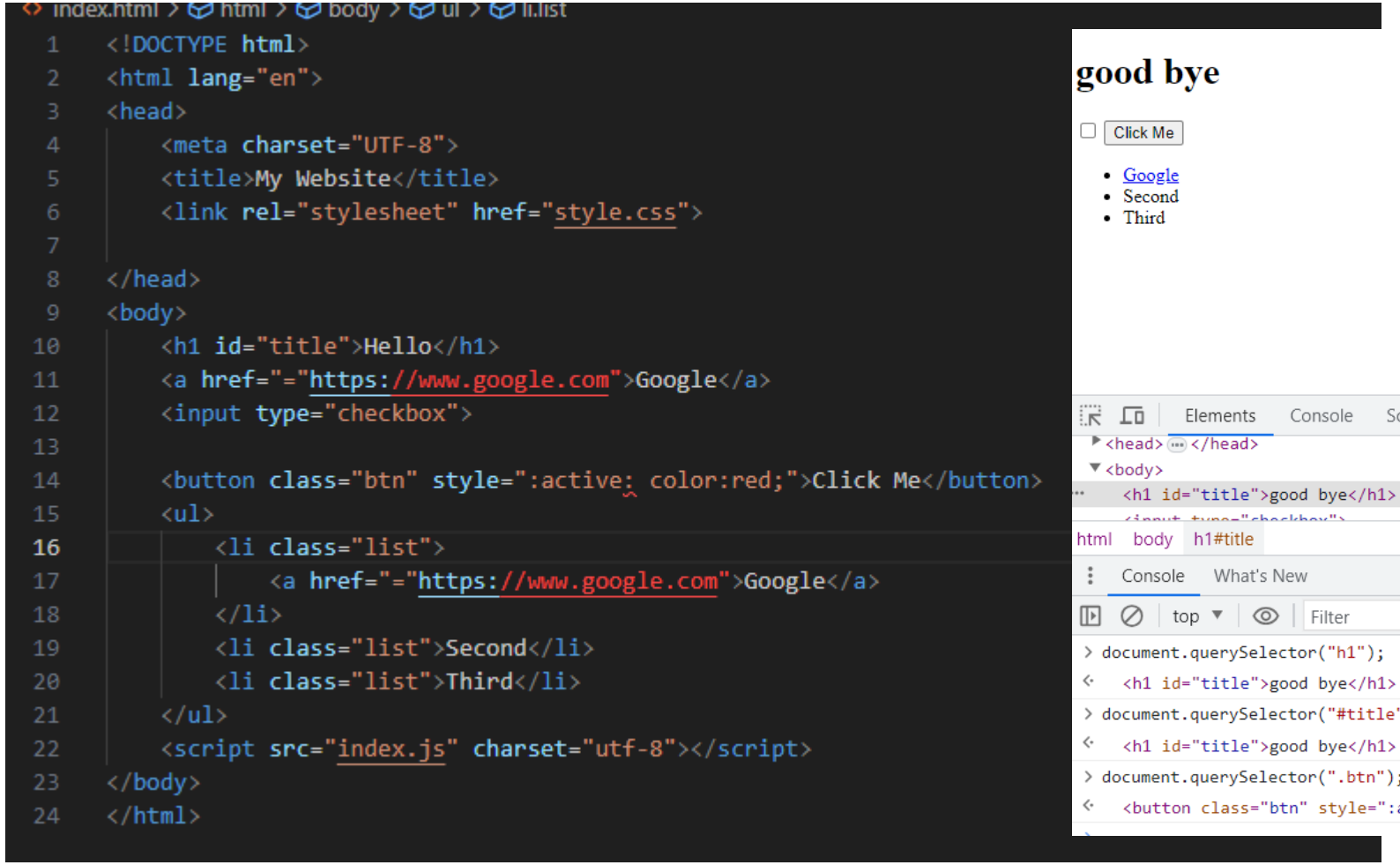
getElementByClassName



getElementById



querySelector

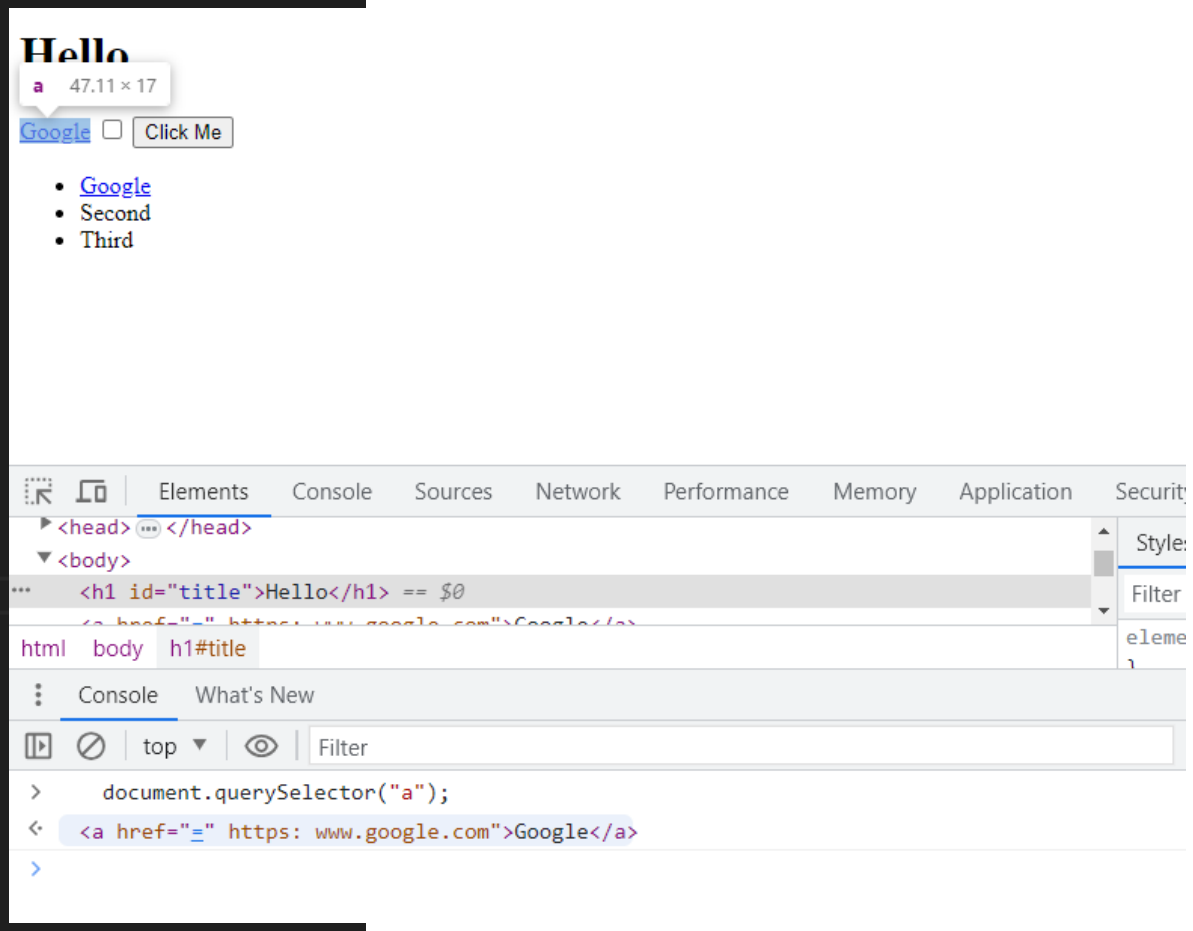


index.html > html > body > ul > li.list

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>My Website</title>
    <link rel="stylesheet" href="style.css">

</head>
<body>
    <h1 id="title">Hello</h1>
    <a href="="https://www.google.com">Google</a>
    <input type="checkbox">

    <button class="btn" style=":active; color:red;">Click Me</button>
    <ul>
        <li class="list">
            <a href="="https://www.google.com">Google</a>
        </li>
        <li class="list">Second</li>
        <li class="list">Third</li>
    </ul>
    <script src="index.js" charset="utf-8"></script>
</body>
</html>
```

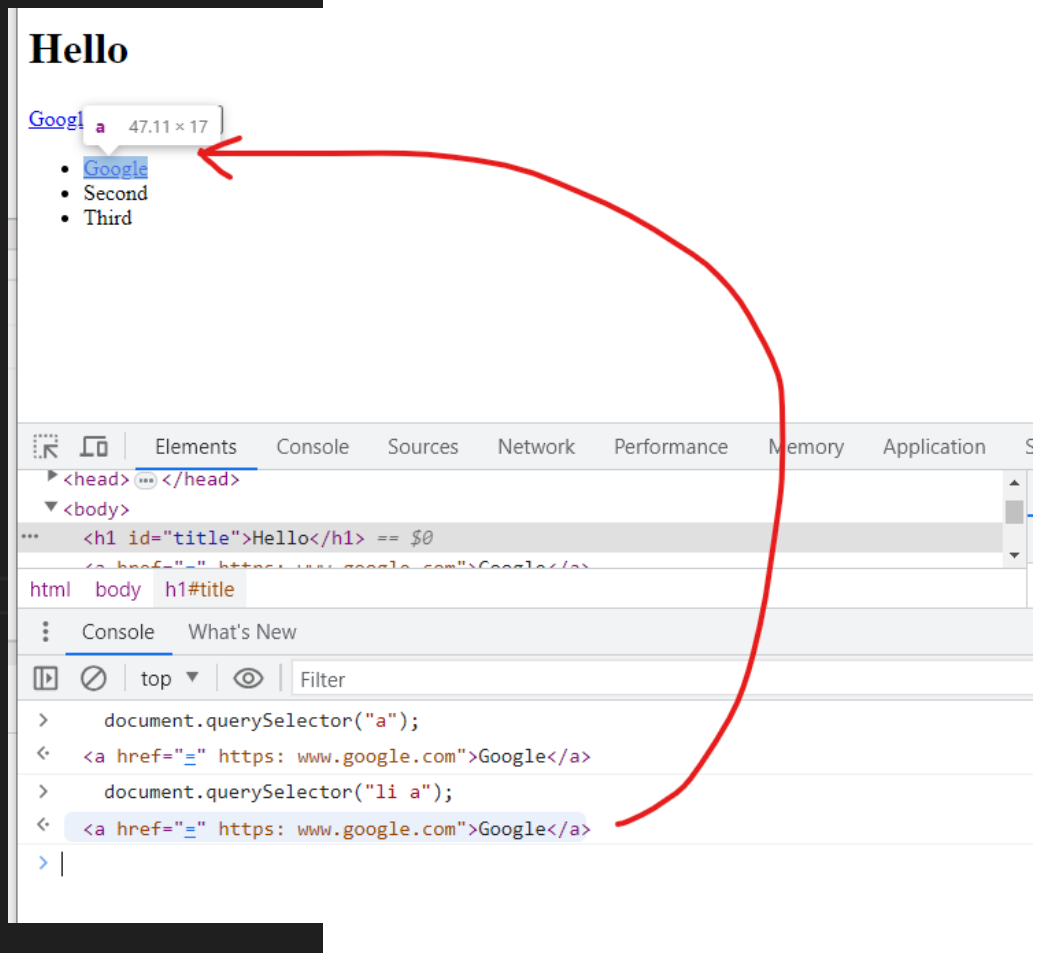


index.html > html > body > ul > li.list

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>My Website</title>
    <link rel="stylesheet" href="style.css">

</head>
<body>
    <h1 id="title">Hello</h1>
    <a href="="https://www.google.com">Google</a>
    <input type="checkbox">

    <button class="btn" style=":active; color:red;">Click Me</button>
    <ul>
        <li class="list">
            <a href="="https://www.google.com">Google</a>
        </li>
        <li class="list">Second</li>
        <li class="list">Third</li>
    </ul>
    <script src="index.js" charset="utf-8"></script>
</body>
</html>
```



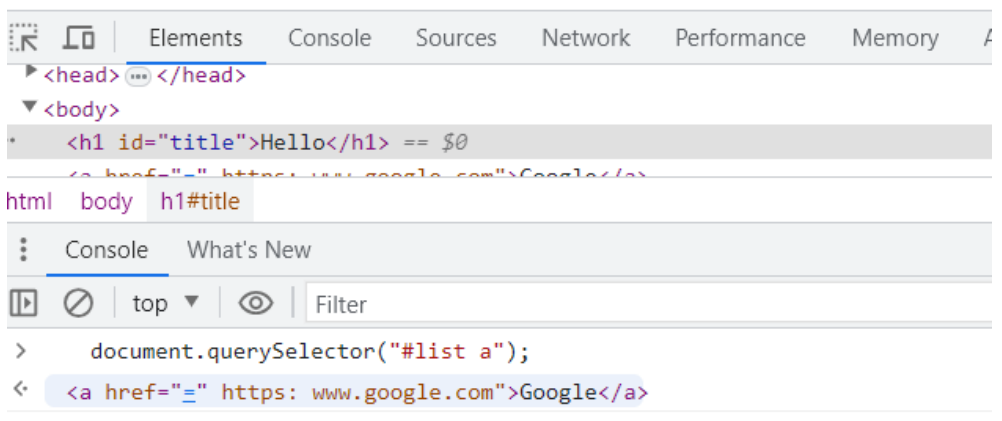
index.html > html > body > ul#list > li.item

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>My Website</title>
    <link rel="stylesheet" href="style.css">

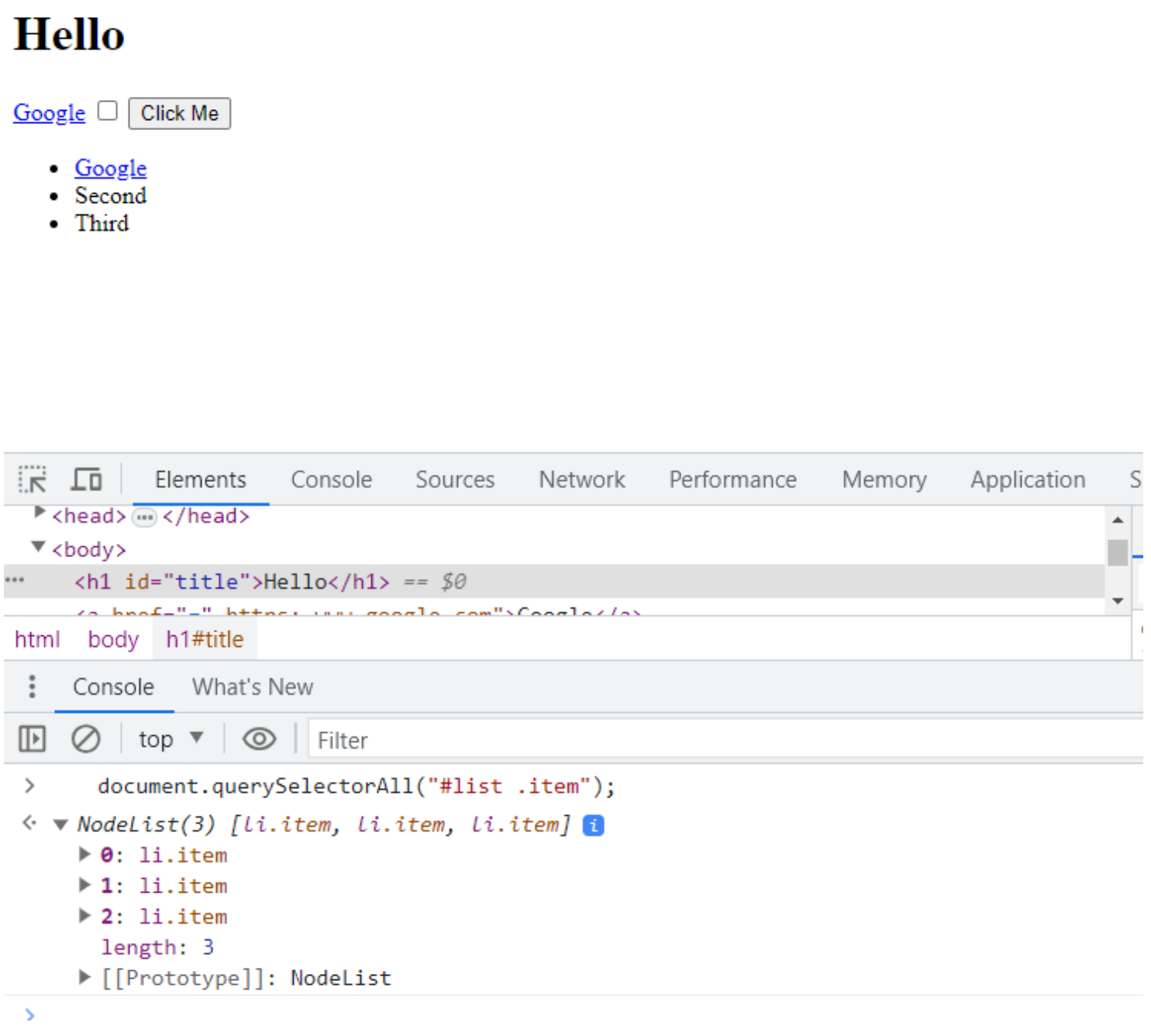
</head>
<body>
    <h1 id="title">Hello</h1>
    <a href="="https://www.google.com">Google</a>
    <input type="checkbox">

    <button class="btn" style=":active; color:red;">Click Me</button>
    <ul id="list">
        <li class="item">
            <a href="="https://www.google.com">Google</a>
        </li>
        <li class="item">Second</li>
        <li class="item">Third</li>
    </ul>
    <script src="index.js" charset="utf-8"></script>
</body>
</html>
```





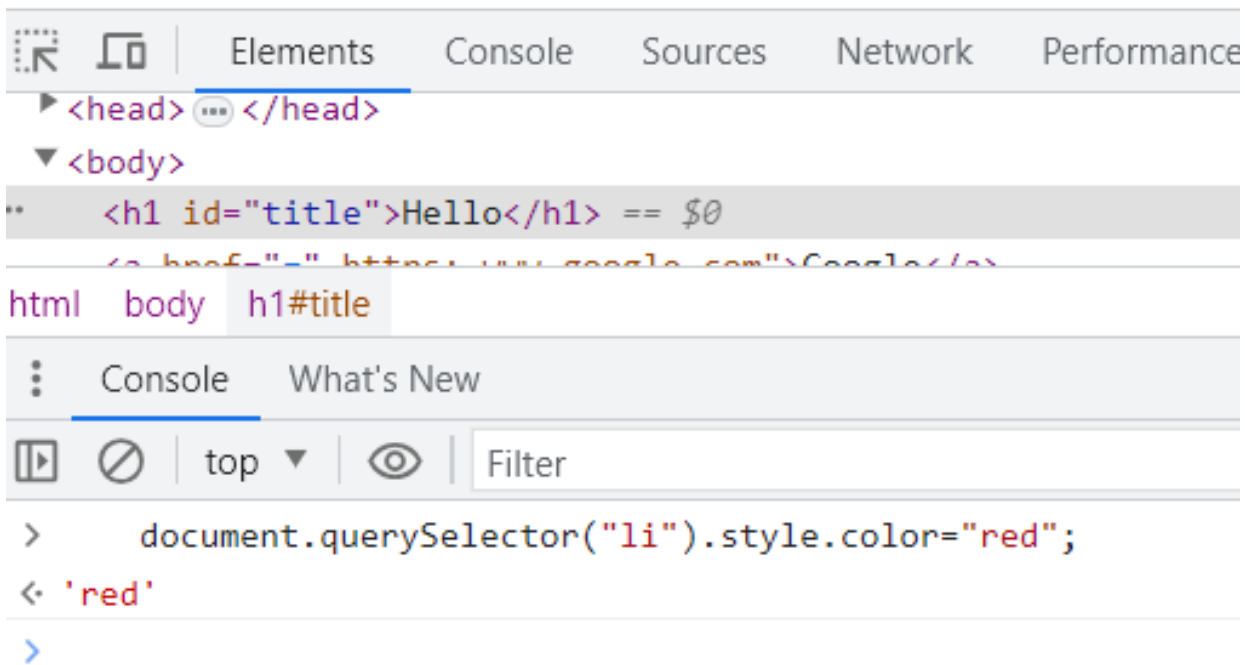
getSelectorAll



Hello

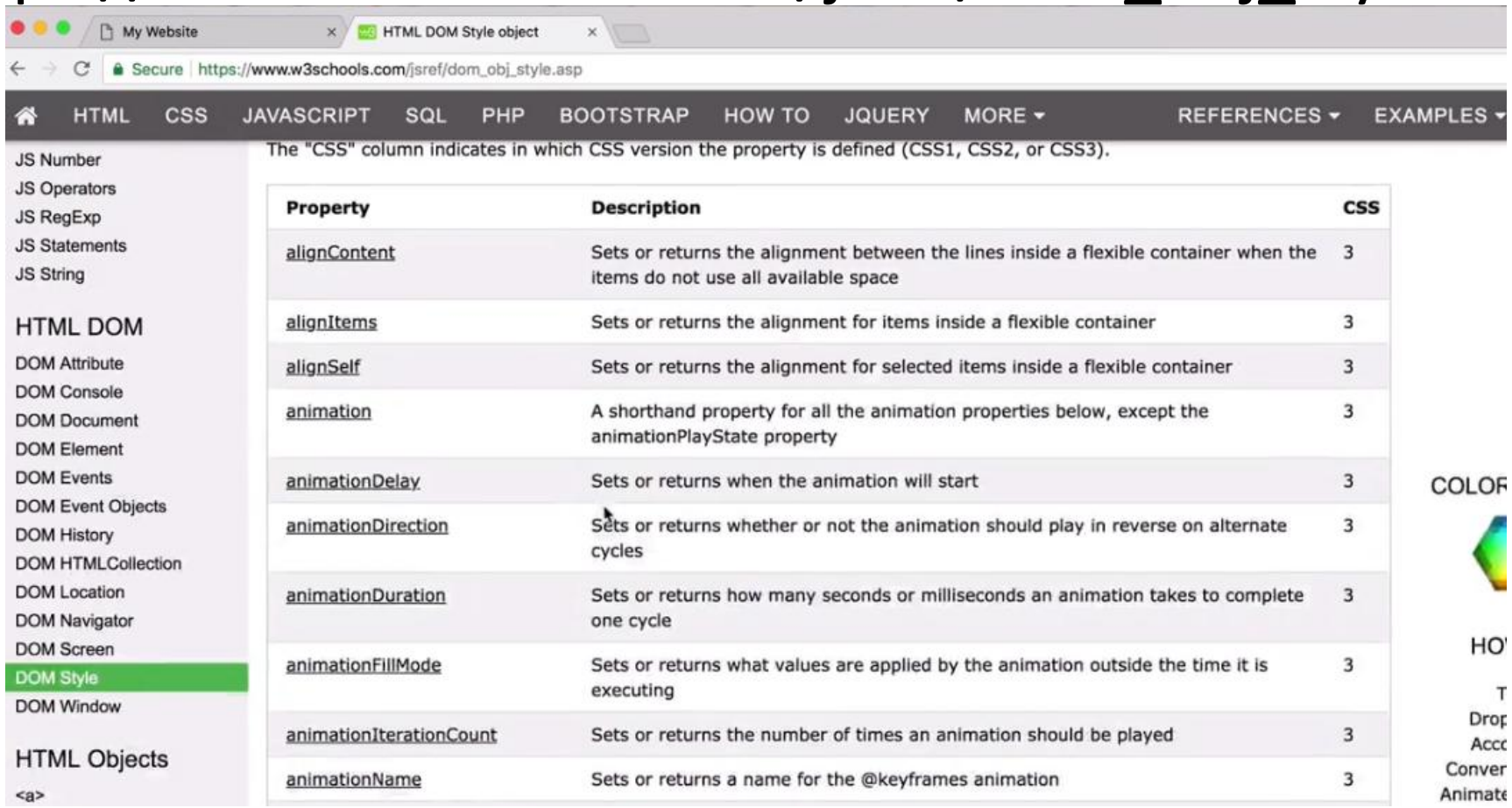
Google Click Me

- Google
- Second
- Third

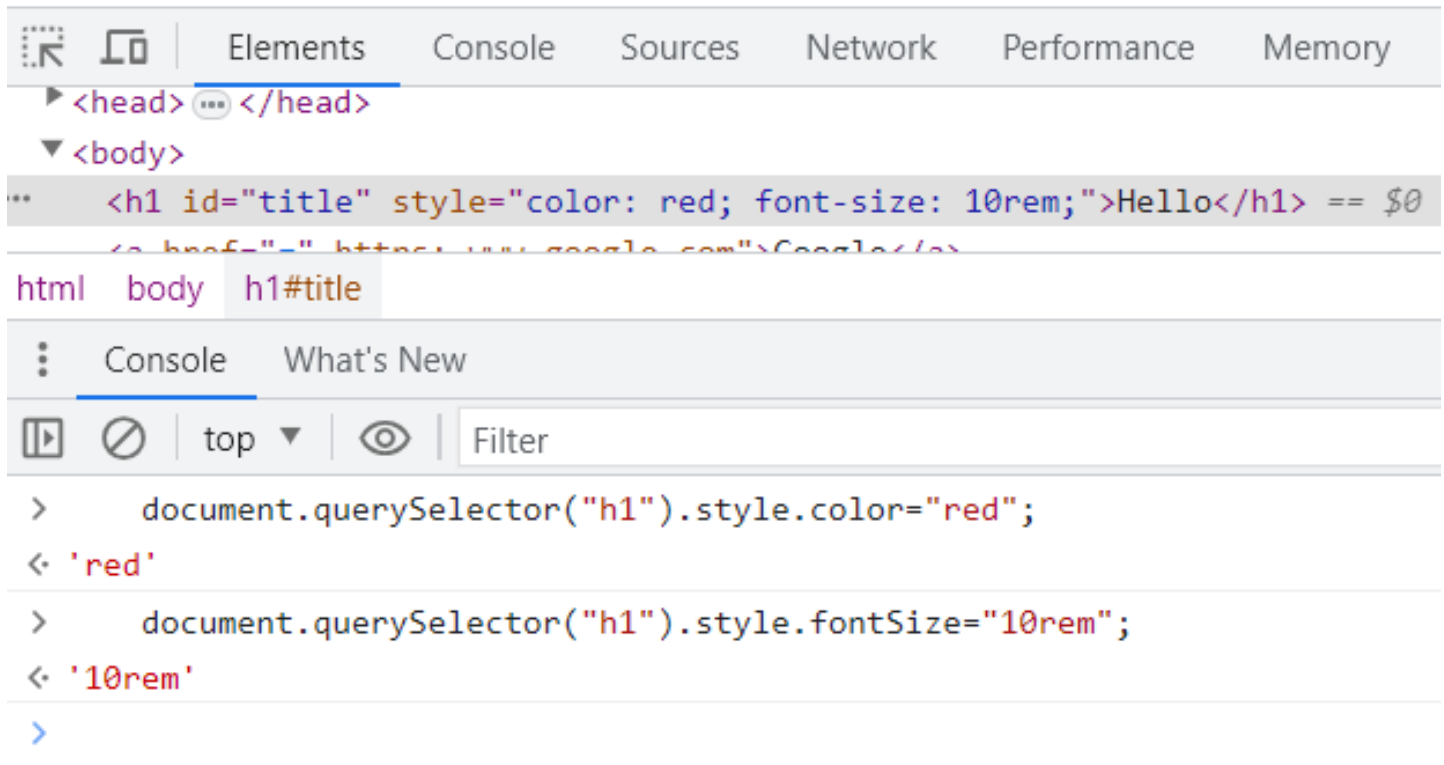


Manipulating and Changing Style

https://www.w3schools.com/jsref/dom_obj_style.asp



Hello


Elements
Console
Sources
Network
Performance
Memory
<head> </head>
<body>
<h1 id="title" style="color: red; font-size: 10rem;">Hello</h1> == $0
html
body
h1#title
Console
What's New
top
Filter
document.querySelector("h1").style.color="red";
'red'
document.querySelector("h1").style.fontSize="10rem";
'10rem'

Style visibility

- document.getElementById("myP").style.visibility = "hidden";

Syntax

Return the visibility property:

```
object.style.visibility
```

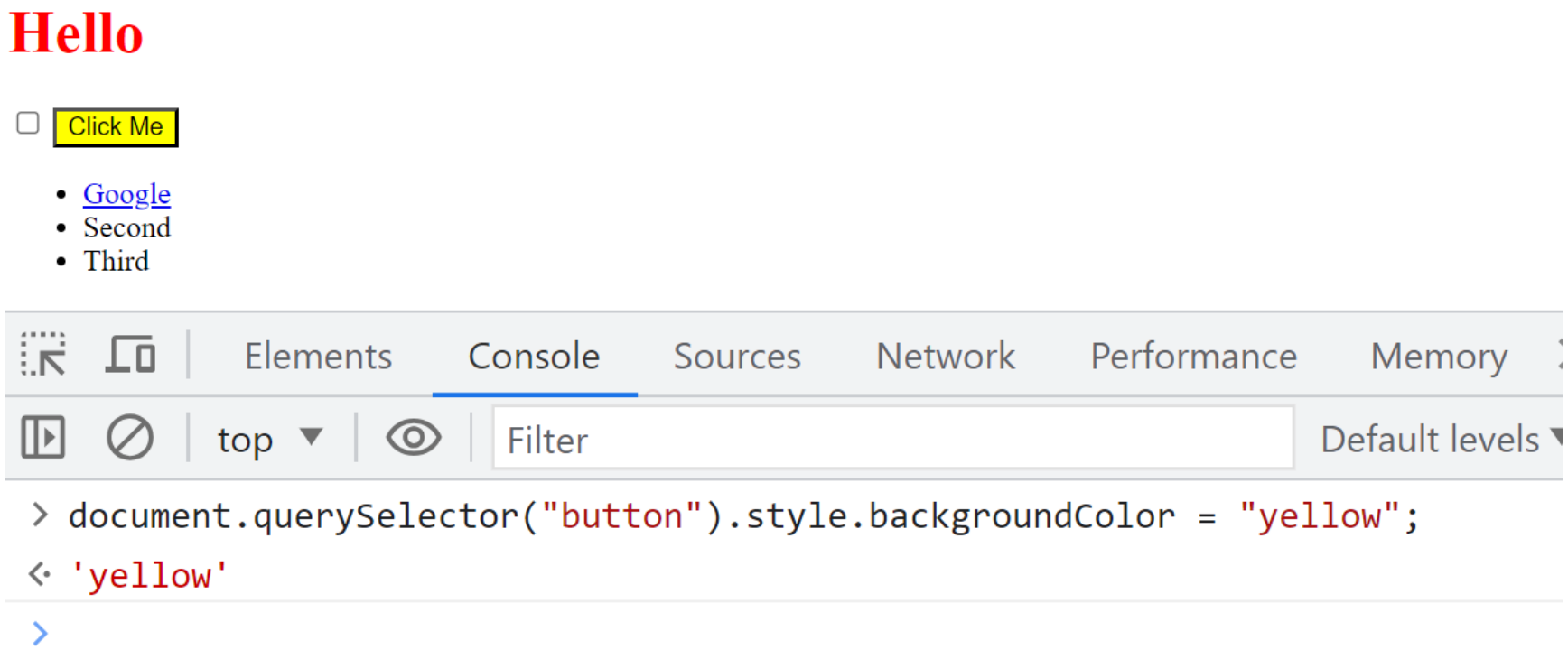
Set the visibility property:

```
object.style.visibility = "visible|hidden|collapse|initial|inherit"
```

Property Values

Value	Description
visible	The element is visible. This is default
hidden	The element is not visible, but still affects layout
collapse	When used on a table row or cell, the element is not visible (same as "hidden")
initial	Sets this property to its default value. Read about *initial*
inherit	Inherits this property from its parent element. Read about *inherit*

Change style by javascript



```html
<!DOCTYPE html>
<html lang="en" dir="ltr">
  <head>
    <meta charset="utf-8">
    <title>My Website</title>
    <link rel="stylesheet" href="styles.css">

  </head>
  <body>

    <h1 id="title"><strong>Hello</strong></h1>

    <input type="checkbox">

    <button style=":active color:red;">Click Me

    <ul>
      <li class="list">
        <a href="https://www.google.com">Google
      </li>
      <li class="list">Second</li>
      <li class="list">Third</li>
    </ul>

  </body>

</html>
```

Hello

Click Me

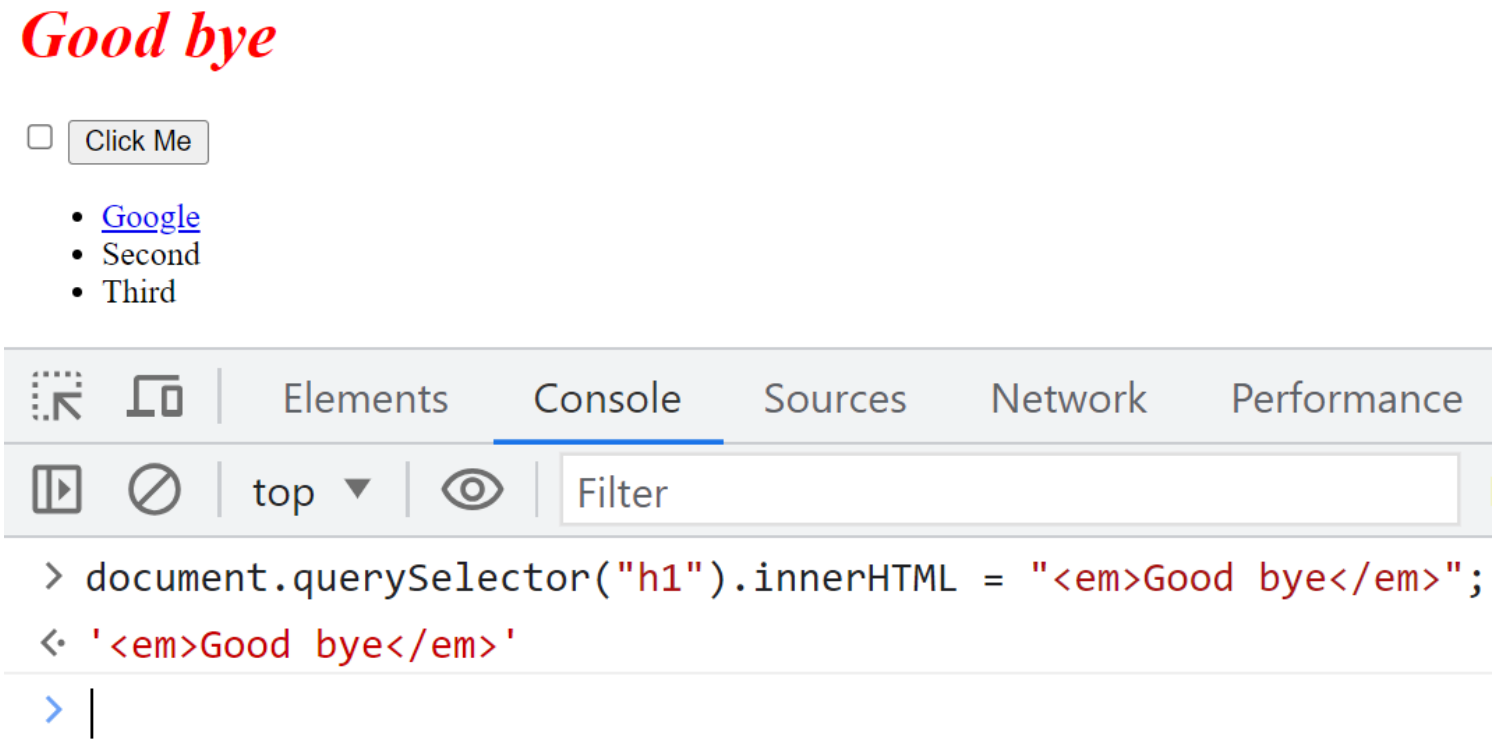
- Google
- Second
- Third

Elements Console Sources Network

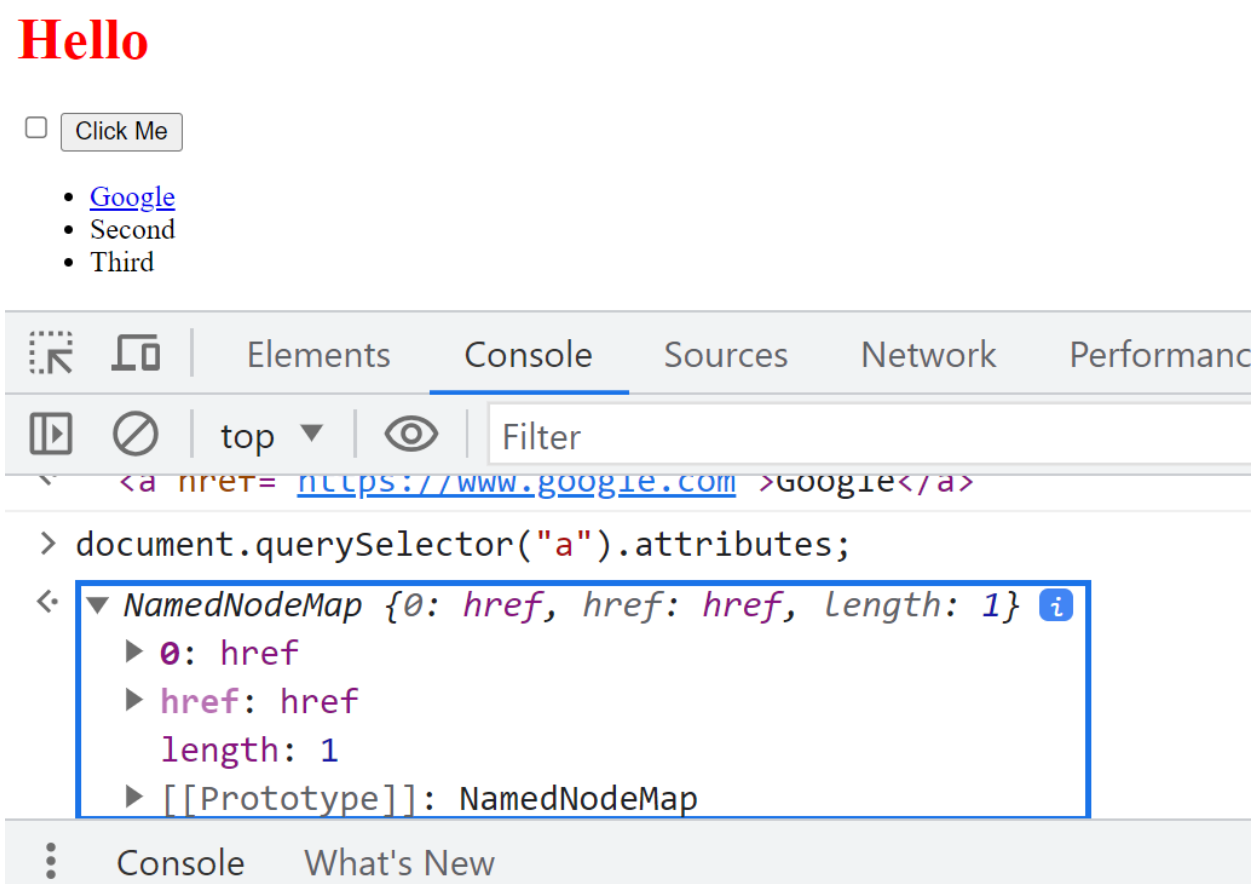
top Filter

```
> document.getElementById("title").innerHTML;
< '<strong>Hello</strong>'
> document.getElementById("title").textContent;
< 'Hello'
>
```

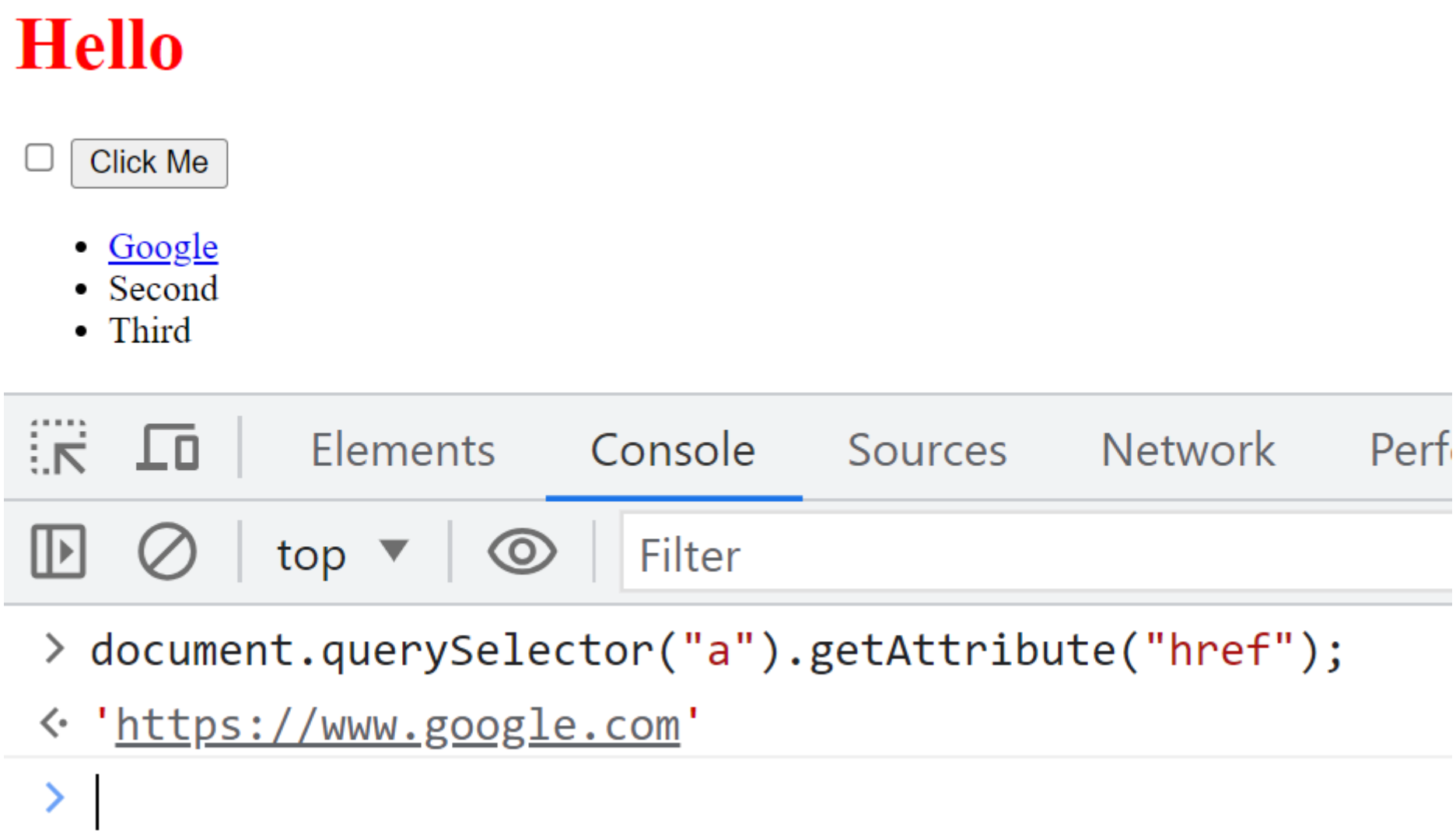
Use HTML in Javascript



Manipulate HTML element attributes



getAttribute



setAttribute

